

$$A^T A = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 6 & -3 & -3 \\ 0 & -3 & 6 & -3 \\ 0 & -3 & -3 & 6 \end{pmatrix}$$

$$\begin{vmatrix} 3-\lambda & 0 & 0 & 0 \\ 0 & 6-\lambda & -3 & -3 \\ 0 & -3 & 6-\lambda & -3 \\ 0 & -3 & -3 & 6-\lambda \end{vmatrix} = (3-\lambda) \begin{vmatrix} 6-\lambda & -3 & -3 \\ -3 & 6-\lambda & -3 \\ -3 & -3 & 6-\lambda \end{vmatrix} =$$

Charakteristický polynom:

$$(3-\lambda)[(6-\lambda)^3 - 27 - 27 - 3 \cdot 9(6-\lambda)] = 0$$

$$(3-\lambda)\lambda(\lambda-9)^2 = 0$$

at

$$\lambda_1 = 0 \quad \text{Ker } M_0 = \text{Ker} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 6 & -3 & -3 \\ 0 & -3 & 6 & -3 \\ 0 & -3 & -3 & 6 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\lambda_2 = 3 \quad \text{Ker } M_3 = \text{Ker} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & -3 & -3 \\ 0 & -3 & 3 & -3 \\ 0 & -3 & -3 & 3 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\lambda_3 = 9 \quad \text{Ker } M_9 = \text{Ker} \begin{pmatrix} -6 & 0 & 0 & 0 \\ 0 & -3 & -3 & -3 \\ 0 & -3 & -3 & -3 \\ 0 & -3 & -3 & -3 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ -2 \end{pmatrix} \right\} = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} \perp \begin{pmatrix} 0 \\ 1 \\ 1 \\ -2 \end{pmatrix}$$

Singularní hodnoty:

$$\sigma_1 = 3$$

$$\sigma_2 = 3$$

$$\sigma_3 = \sqrt{3}$$

$$D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$V^T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$U = \left(\frac{1}{\sigma_1} A v_1 \mid \frac{1}{\sigma_2} A v_2 \mid \frac{1}{\sigma_3} A v_3 \right) = \begin{pmatrix} 0 & \frac{\sqrt{6}}{3} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A = U D V^T$$

$$A^T = (UDV^T)^T = VD^T U^T$$

$A^T A$ je symetrická, proto $A^T A = (A^T A)^T$.

Dále $A^T A (A^T A)^T = (A^T A)^2$. ~~Tedy~~ Takže vlastní čísla $(A^T A)^2$ jsou $\lambda_1^2 = 0, \lambda_2^2 = 9, \lambda_3^2 = 81$ vlastní vektory stejné jako u $A^T A$. Singulární hodnoty jsou $\sigma_1 = 9, \sigma_2 = 9, \sigma_3 = 3$

$$U' = \left(\frac{1}{\sigma_1} A^T A v_1 \mid \frac{1}{\sigma_2} A^T A v_2 \mid \frac{1}{\sigma_3} A^T A v_3 \right) = \begin{pmatrix} 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$D' = \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

$$A^T A = U' D' V^T$$

$$\begin{aligned} A A^T &= U D V^T V D^T U^T = \\ &= U D D^T U^T \\ &= U \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix} U^T \end{aligned}$$